

Known Limitations Documentation

Architectural Boundaries, Mocks, Simplifications, and Production Roadmap

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1. Operational Boundaries & Classification Taxonomy

In accordance with Section 77 of the PRD, this document explicitly discloses what capabilities are fully implemented in production architecture, what elements are simulated via deterministic mocks, what areas are simplified, and what remains on the production roadmap:

FULLY IMPLEMENTED (Production Architecture)

- **Multi-Tenant Server Isolation:** Strict tenant separation at database, repository, service, worker, and RAG levels.
- **Four-Tier Role-Based Access Control:** PLATFORM_ADMIN, HOSPITAL_ADMIN, CAMPAIGN_MANAGER, CLINICAL_REVIEWER route guards.
- **Capacity-Aware Priority Queue:** Concurrency line governor, deadline pressure exponentiation, starvation aging factor.
- **Dual AI Clinical Assessments:** Independent LLM Assessment A paired with deterministic Rule Validator Assessment B and Conservative Arbiter.
- **Human-in-the-Loop Review Desk:** Clinical reviewer inbox with resolution actions, note logging, and EHR synchronization.
- **Append-Only Audit Logging:** Immutable event stream recording state transitions, actor identities, and correlation IDs.

SIMULATED (High-Fidelity Deterministic Mocks)

- **Telephony Carrier (SIP / WebRTC Trunk):** Live Twilio voice connection is replaced by a deterministic telephony simulator supporting all outcomes (NO_ANSWER, BUSY, VOICEMAIL, DROPPED, CALLBACK_REQUESTED).
- **Patient Speech Interaction:** Real-time human patient voice is simulated via 7 diverse conversational personas (Routine, Urgent, Concerning, Ambiguous, Incomplete, Prompt Injection).
- **EHR Vendor Backend:** The MockEHRService simulates FHIR/HL7 REST API interactions with configurable latency and error recovery modes.

SIMPLIFIED FOR OPERATIONAL PROTOTYPE

- **FHIR Resource Depth:** Implements core post-discharge resources (Patient, Encounter, Discharge, Condition, Observation) rather than full 150+ FHIR R4 schema specifications.
- **Single-Node SQLite Storage:** Configured in high-performance Write-Ahead Logging (WAL) mode with mutex locking for zero-dependency local execution, alongside full PostgreSQL schema definitions.

2. Production Roadmap & Scalability Enhancements

The following engineering initiatives transition the operational prototype to national-scale multi-hospital deployment:

- **Live Telephony Carrier Integration:** Replace the telephony simulator with Twilio Voice / Amazon Connect using Media Streams + WebSockets for sub-500ms voice turn-taking.
- **Distributed Message Queue (Redis Streams / Celery):** Migrate the in-process queue scheduler to distributed Redis Streams, supporting thousands of concurrent worker pods across multi-region clusters.
- **SMART on FHIR mTLS Interoperability:** Implement OAuth2 SMART on FHIR app launch and mTLS HL7 v2 messaging for bi-directional synchronization with Epic Hyperspace and Cerner Millennium.
- **Multilingual ASR & Dialect Normalization:** Expand conversational acoustic models to support Spanish, Mandarin, and regional accents with automated dialect normalization.

Clinical Safety Assurance: While telephony and live EHR connectivity are simulated for evaluation safety, **all core clinical safety algorithms (concurrency locking, conservative consensus arbiter, red-flag escalation, tenant boundaries) are fully implemented in real production code** and backed by 100% passing automated test suites.